

Deployment Guide

Customer Segmentation Dashboard

1. Local Deployment

1. Ensure Python 3.10+ is installed.
2. Clone or copy the customer-segmentation-project/ folder.
3. From the project root, install dependencies: `pip install -r requirements.txt`
4. Run: `streamlit run app/app.py`
5. Open the URL shown in the terminal (default `http://localhost:8501`).

2. Environment Setup

No .env file or secrets are required for local deployment. A virtual environment is recommended:

```
python -m venv venv source venv/bin/activate # Windows: venv\Scripts\activate pip
install -r requirements.txt
```

3. Required Files

File / Folder	Required For
data/processed/customer_segments_with_strategy.csv	All dashboard sections (main data source)
models/production_pipeline.pkl	Live prediction (Section 9, "Analyze New Customer")
src/*.py	Shared preprocessing / feature engineering / prediction logic
app/*.py, app/components/*.py	The dashboard application itself

4. Dataset Requirements

The processed CSV must contain the Segment_Name, Cluster, and all 12 clustering-feature columns (Customer_Age, Income, Total_Spending, Recency, Customer_Tenure, Family_Size, Total_Children, Total_Purchases, Total_Campaign_Acceptance, NumWebPurchases, NumStorePurchases, NumCatalogPurchases) plus the display fields used across the 10 sections (Education, Marital_Status, MntWines etc.).

5. Model Files

models/production_pipeline.pkl is a joblib-pickled dict with keys feature_list, scaler, model, k, and random_state. It must be regenerated (via the Module 06 clustering_pipeline.py) if the underlying dataset or feature set changes.

6. Online Deployment (Optional)

For cloud deployment, Streamlit Community Cloud is the simplest option: push the repository to GitHub, connect it at share.streamlit.io, and set the app entry point to app/app.py. Ensure data/ and models/ are included in the repository (or fetched at startup from external storage) since the app depends on them at runtime. For production traffic, consider containerizing with Docker and deploying to a cloud provider (AWS/Azure/GCP) behind a load balancer, with the dataset served from a managed database rather than a static CSV.

7. Post-Deployment Checklist

- Confirm the dashboard loads without errors and all 10 sections render.
- Confirm filters update charts/tables live.
- Confirm a known Customer ID returns the expected segment.

- Confirm the "Analyze New Customer" form produces a sensible prediction and rejects invalid input with a clear message.
- Run the pytest suite (pytest tests/ -v) in the deployment environment to confirm all 40 tests still pass.